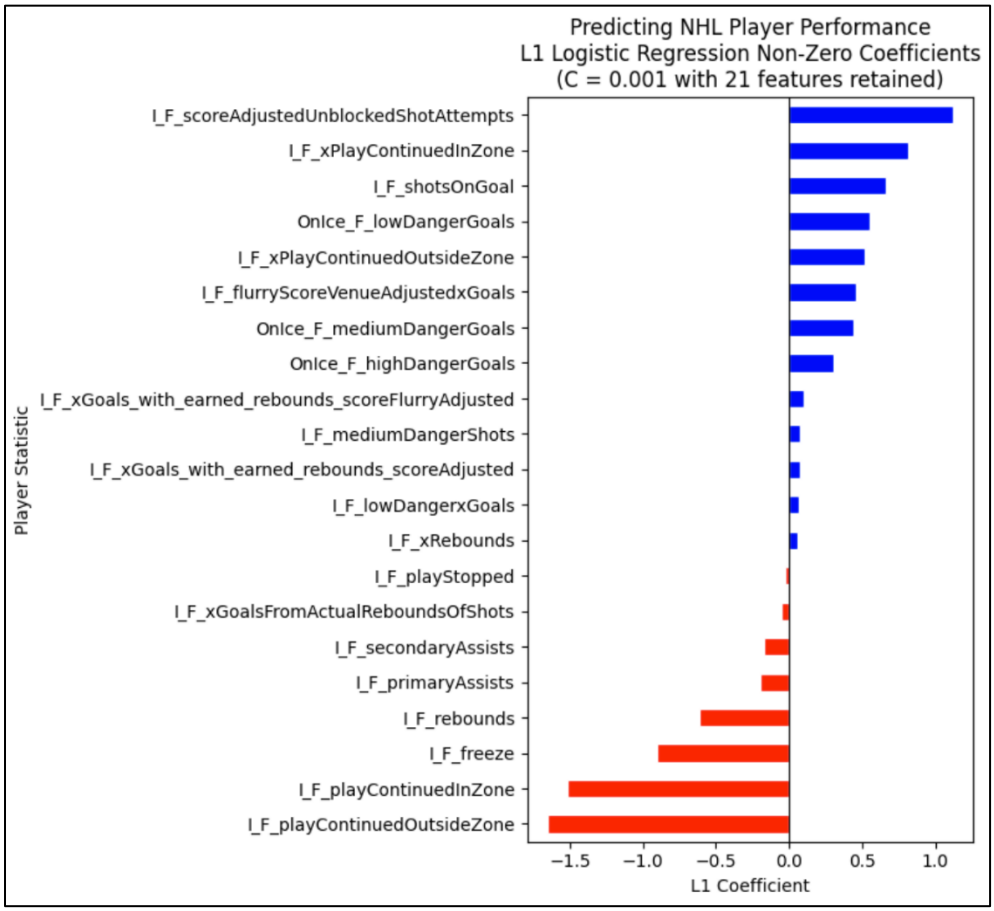
Modelling NHL Player Performance Metrics



Key Findings

- A player's ability to find clear ("unblocked") shots through plays that preserve momentum serves the strongest predictor for that player's overperformance during an NHL game.
    - Score-adjusted unblocked shot attempts was the strongest overperformance signal (L1 coef: +1.12), followed by expected play continued in zone (+0.82) and shots on goal (+0.66). On-ice low/medium/high danger goals also survived, suggesting team offensive context matters.
    - Play continued outside the zone (−1.65) and play continued in zone (−1.52) were the strongest negatives, along with freeze events (−0.90) and rebounds (−0.61).

These reflect situations where a shot occurred without any significant chances generated, either due to poor shot quality or shot selection.

- L1 regularization was able to effectively capture signal from a high-dimensional dataset (21 features preserved out of 138 quantitative contenders) and meaningfully predict a player's overperformance in a classification setting through a logistic regression model.

Data & Methods

National Hockey League game-level player data was collected from MoneyPuck for the 2024-2025 season, the most recent complete NHL season. A player's goals-above-expected (`goalsAboveExpected') was calculated by differencing a player's game goal total ('I_F_goals') and their expected goals ('I_F_xGoals'), with the latter being an advanced statistic intended to serve as a prediction for the player's goal production during the game. A player can therefore be "overperforming" during any given game if they have a positive goals-above-expected metric (goalsAboveExpected > 0) by scoring more goals than their expected goal count. A binary target variable was created to map a player's performance to either an "overperformance" or an "underperformance" using the calculated metric. This motivates a clear classification problem to explore the driving factors behind player overperformance.

Given that goals are a relative rarity in the NHL and there are a number players are listed in the dataset that did not play in a given game (caused by scratches, game-time injuries, etc.), a strict requirement is enforced that any valid player data entry must have at least 1 shift logged for the game (shifts > 0). This approach avoids the significant noise provided by data entries without meaningful statistics from which signal can be extracted.

Prior to prediction, certain variables were removed from the possible set of predictive features due to a high likelihood of information leakage. Given the binary response variable and the high dimensionality of the dataset, a logistic regression equipped with L1 regularization was selected for the prediction of a player's performance. L1 regularization was selected to shrink the coefficients of low variance features to zero, ensuring that any remaining features held significant predictive power. This would improve the interpretability of the logistic regression model without further limiting the pool of predictors.

5-fold cross validation was utilized to tune lambda, the L1 hyperparameter, with 20% of the original dataset held out for model evaluation. Some analyst discretion through the use of the elbow method was applied in selecting the optimal lambda to preserve some regularizing ability of the L1 regression. ROC-AUC was used as the evaluation metric for the models trained during 5-fold cross validation due to its relevance in a classification setting. After an appropriate hyperparameter was determined, a final, tuned logistic regression model was fit to the training data and evaluated against the holdout test data.

While the purpose for the L1 logistic regression model was to prioritize interpretability over predictive accuracy, the model trained on advanced statistics yield an accuracy of 96.34%, a notable improvement over the naïve baseline of roughly 91.72%, validating the efficacy of the approach.

Relevance of Findings

Sports analytics stands as one of the most widely explored, culturally relevant forms of data analytics in the modern age. Interpreting the relevant factors that contribute to player performance allows players and coaches to optimize training, game strategy, and player load management throughout the course of a season. Hockey is a particularly dynamic sport with multiple defining swings in overall strategy and player skill prioritization throughout its lifespan, underscoring the need for in-depth player analysis beyond direct scoring metrics. The findings of this report suggest that the current league prioritization of shooting talent is warranted given the strong positive association between shot quality/frequency and incremental performance over an individual player's predicted performance.

Link to Accompanying GitHub Repository

https://github.com/epdeignan/INFSCI1530-Final-Report.git